60 single-best-answer questions | 90 minutes | 60 marks

Answer all questions. Choose one answer per question. Use a separate answer sheet.

1. A patient has a sudden rise in arteriolar sympathetic firing with minimal change in cardiac output. Which immediate variable rises most directly?

A. The intrinsic rate of the SA node.
B. The rate of tissue metabolite clearance.
C. Total peripheral resistance.
D. Systemic venous compliance.

2. The R-R separation is four large squares on a regular standard-speed ECG. What is the heart rate?

A. 75 beats/min.
B. 120 beats/min.
C. 60 beats/min.
D. 90 beats/min.

3. An ECG develops a new shift of the ST segment above or below the baseline. Which process specifically makes this segment clinically important?

A. Myocardial ischaemia or inadequate oxygenation can alter ST level.
B. Normal atrial depolarisation creates the ST deflection of each beat.
C. A decrease in cardiac preload directly determines ST segment height.
D. The onset of valve closure normally lifts the ST segment above baseline.

4. Portal hypertension promotes ascites partly by increasing which force?

A. Plasma oncotic pressure within hepatic capillaries.
B. Hydrostatic pressure in hepatic sinusoids.
C. Interstitial lymphatic pumping efficiency.
D. Arterial oxygen tension in the hepatic vein.

5. Which contrast best distinguishes forward failure from backward failure?

A. Forward reduces delivery; backward causes venous congestion.
B. Forward is venous congestion; backward is low organ blood flow.
C. Forward indicates excess cardiac output; backward low afterload.
D. Forward is confined to the right heart; backward to the left.

6. After severe head injury, a patient has high blood pressure, a slow pulse and irregular breathing. What does this combination suggest?

A. An uncomplicated rise in exercise-related cardiac output and flow.
B. A dangerous Cushing response to intracranial compression.
C. A typical active-hyperaemia response in skeletal muscle.
D. An isolated atrial volume reflex after rapid infusion.

7. What directly promotes pulmonary oedema in left-sided backward heart failure?

A. Reduced pulmonary capillary hydrostatic pressure.
B. Reduced systemic venous pressure throughout the body.
C. Raised pulmonary venous and capillary pressure.
D. Raised right atrial pressure without lung congestion.

8. Why does the fetal cardiovascular system divert much blood away from the pulmonary circuit before birth?

A. The fetal left ventricle cannot generate arterial pressure.
B. Pulmonary vascular resistance is exceptionally low before birth.
C. Umbilical arteries carry already oxygenated blood to the heart.
D. The placenta oxygenates blood; fetal lungs resist flow.

9. After bleeding stops, both haemoglobin and plasma-protein concentrations fall as circulating volume recovers. What explains this?

A. Added fluid dilutes the remaining blood components.
B. New red-cell production lowers haemoglobin concentration.
C. Raised renal filtration removes most circulating proteins.
D. Venoconstriction selectively removes haemoglobin from blood.

10. An ECG electrode detects a voltage on the skin even though the impulse originates in the myocardium. What makes this possible?

A. The heart mechanically transmits its pulse to every skin electrode.
B. The electrodes measure changes in circulating blood pressure.
C. Cardiac electrical currents spread through conducting body tissues.
D. The chest electrodes directly contact ventricular muscle fibres.

11. Heart rate rises from 70 to 80 beats/min while stroke volume stays at 70 mL. By how much does cardiac output increase?

A. By 0.80 litres per minute.
B. By 0.70 litres per minute.
C. By about 0.49 litres per minute.
D. By 1.40 litres per minute.

12. At unchanged blood density, vessel diameter and viscosity, what happens to Reynolds number if average blood velocity doubles?

A. A twofold increase.
B. A twofold decrease.
C. A fourfold increase.
D. No change in value.

13. Most sympathetic constrictor fibres release noradrenaline onto arteriolar smooth muscle. Which receptor mainly mediates constriction?

A. The beta-2 adrenergic receptor on vessels.
B. The muscarinic M2 receptor.
C. The endothelial H2 receptor.
D. The alpha-1 adrenergic receptor.

14. A harsh systolic murmur develops during blood ejection through a narrowed aortic valve. What mechanism explains it?

A. Opening of the aortic valve during ventricular relaxation.
B. Turbulent forward flow through a stenotic outflow orifice.
C. Regurgitation through the mitral valve during diastole.
D. Laminar blood flow through a normal-sized valve.

15. What distinguishes the normal x descent from the y descent in a JVP trace?

A. x: atrial relaxation; y: emptying through the open tricuspid.
B. x: atrial contraction; y: ventricular isovolumetric contraction.
C. x: ventricular ejection; y: pulmonary arterial recoil.
D. x: aortic valve opening; y: mitral valve closure.

16. At 25 mm/s, a PR interval extends across 4.5 small squares. Which interpretation is most accurate?

A. 0.24 s, above the stated PR-interval range.
B. 0.09 s, shorter than the stated normal PR-interval range.
C. 0.90 s, above the stated PR-interval range.
D. 0.18 s, inside the stated PR-interval range.

17. Why does fetal haemoglobin load oxygen effectively at the placental interface?

A. It binds 2,3-BPG more tightly than adult haemoglobin.
B. Its weaker 2,3-BPG binding increases oxygen affinity.
C. It is produced without haem groups before birth.
D. Its iron permanently binds each oxygen molecule.

18. A patient develops headache that is worse upright after lumbar CSF sampling. Which mechanism is most likely?

A. Persistent CSF leakage lowers intracranial CSF pressure.
B. The sample immediately increases blood-brain barrier permeability.
C. Lumbar puncture increases CSF production several-fold.
D. The procedure obstructs both foramina of Monro.

19. During severe hepatic hypoperfusion, cells near the central vein are especially vulnerable to hypoxia. Why?

A. They are located outside the hepatic sinusoidal network.
B. They are closest to arterial inflow.
C. They are farthest downstream from the portal triads.
D. They receive blood with the highest oxygen content.

20. Following a substantial meal, intestinal blood flow rises for several hours. What primarily drives this response?

A. Global sympathetic constriction of intestinal arteries.
B. Reduced intestinal metabolism during nutrient absorption.
C. Complete diversion of portal blood away from liver.
D. Local metabolic activity and absorptive demand.

21. Two identical limbs have arterial flow interrupted for one minute and five minutes respectively. Which reperfusion pattern is expected?

A. Neither block produces an overshoot after release.
B. The five-minute block produces a larger, longer surge.
C. Both blocks produce identical flow responses in time.
D. The one-minute block produces a larger and more lasting surge.

22. Two Frank-Starling curves are plotted against identical end-diastolic volumes. What does an upward shift of the entire curve usually indicate?

A. Greater contractility at the same preload.
B. Greater afterload alone with unchanged contractility.
C. Aortic valve closure earlier in systole.
D. Lower force development at every fibre length.

23. During a stress response, circulating adrenaline dilates vessels in working skeletal muscle while constricting many skin vessels. Why?

A. The muscle vessels lack any adrenergic receptors.
B. Different vascular beds express different adrenergic receptors.
C. Skin tissue receives a much lower concentration of adrenaline.
D. Different vascular beds contain markedly different amounts of plasma.

24. An older patient faints shortly after standing to urinate at night. Which phenomenon does this episode most closely illustrate?

A. Carotid sinus syncope caused by neck pressure.
B. Micturition syncope linked to reflex hypotension.
C. Septic shock due to widespread inflammatory vasodilatation.
D. Positive-g blackout from head-level underperfusion.

25. In severe haemorrhagic shock, why does sympathetic constriction of splanchnic vessels help initially?

A. It increases flow to the intestinal mucosa above baseline.
B. It increases hepatic oxygen extraction to 100%.
C. It prevents all intestinal ischaemia during shock.
D. It redistributes flow and mobilises stored venous blood.

26. An examination reveals raised JVP, a tender enlarged liver, ascites and ankle oedema. Which haemodynamic change unifies these?

A. High systemic venous pressure behind the right ventricle.
B. Reduced filtration from peripheral capillaries.
C. Increased pulmonary venous pressure without systemic congestion.
D. Reduced central venous pressure from depleted blood volume.

27. An isolated SA node fires at about 100 beats/min, whereas the same person's resting pulse was about 70. Which interpretation best fits?

A. Atrial conduction had slowed the spontaneous SA nodal rate.
B. The cardiac pacemaker was intrinsically limited to 70.
C. Resting vagal activity had restrained SA nodal firing.
D. Resting sympathetic activity had restrained SA nodal firing.

28. Immediately after the first effective breaths, what happens to pulmonary vascular resistance?

A. It rises because alveolar oxygen stimulates constriction.
B. It remains unchanged until umbilical vessels close.
C. It equals systemic resistance because both circuits merge.
D. It falls as the lungs expand and oxygen tension rises.

29. A technician places V1 at the fourth intercostal space. Where should the electrode be positioned horizontally?

A. At the left midclavicular line.
B. At the anterior axillary line.
C. At the right sternal border.
D. At the left midaxillary line.

30. Two beats have different QRS heights. A student concludes the taller complex produced a larger stroke volume. What is the main flaw?

A. QRS height represents the duration of ventricular contraction.
B. QRS height directly records the pressure within the ventricle.
C. QRS height measures the amount of venous blood returning.
D. QRS height reflects voltage, not the amount of blood ejected.

31. Which finding would most strongly suggest pulmonary rather than isolated dependent leg oedema?

A. Swelling that worsens around the ankles during standing.
B. Skin indentation remaining after pressing the ankle.
C. A unilateral swollen arm after mastectomy.
D. Lung crackles and dyspnoea with left-heart failure.

32. Why can longstanding filarial lymphoedema become firm and poorly pitting?

A. Chronic protein-rich interstitial fluid promotes tissue fibrosis.
B. Ongoing rapid capillary reabsorption removes tissue proteins.
C. Longstanding venous dilation prevents connective-tissue changes.
D. A brief loss of arterial pressure causes permanent limb scarring.

33. Immediately after rapid loss of a large volume of whole blood, haemoglobin concentration is unexpectedly unchanged. Why?

A. Red-cell mass increases enough to compensate.
B. Plasma enters faster than red cells are lost.
C. Red cells and plasma were lost together.
D. Erythropoietin instantly makes new red cells.

34. A patient with left-sided forward heart failure reports fatigue and produces very little urine. Which mechanism links the two?

A. Elevated venous pressure increases renal filtration.
B. Reduced systemic perfusion limits muscle and renal blood flow.
C. Pulmonary congestion causes systemic arteriolar dilation.
D. Increased left ventricular output stimulates renal water loss.

35. A recording shows that cardiac vagal stimulation lowers the firing rate of the sinoatrial node. Which additional effect is expected?

A. Stronger ventricular contraction through beta-1 activation.
B. Faster impulse conduction through the AV node.
C. Slower impulse conduction through the AV node.
D. Faster pacemaker discharge through alpha-1 activation.

36. A patient has hydrocephalus due to impaired absorption by arachnoid villi without an obstructed ventricular pathway. What type is this?

A. Noncommunicating hydrocephalus.
B. Obstructive aqueductal hydrocephalus.
C. Communicating hydrocephalus.
D. Isolated ventricular haemorrhage.

37. What conservation principle underlies Fick measurement of cardiac output?

A. Venous oxygen usually matches arterial oxygen content.
B. Oxygen consumption is independent of blood flow and extraction.
C. Oxygen use equals flow x arterial-venous oxygen difference.
D. The heart produces all consumed oxygen internally.

38. What keeps tissue fluid moving through collecting lymph vessels toward the great veins?

A. Valves, muscle movement and lymphatic contractions.
B. Direct diffusion of whole lymph vessels through tissues.
C. Constant backflow from the vena cava.
D. Only the left ventricular ejection pressure.

39. An arteriole's radius falls to half its previous value; other variables stay constant. How does its resistance change?

A. By a factor of 2.
B. By a factor of 4.
C. By a factor of 8.
D. A factor of 16.

40. A patient with left-sided heart failure feels breathless when lying flat but improves after sitting up. Why?

A. Lying flat shifts peripheral blood toward congested lungs.
B. Lying flat reduces central blood volume.
C. Sitting up directly increases pulmonary capillary pressure.
D. Sitting up increases venous return from the legs.

41. A regular rhythm has 24 small squares between adjacent R waves at standard paper speed. Approximately what is the heart rate?

A. 50 beats/min.
B. 80 beats/min.
C. 63 beats/min.
D. 60 beats/min.

42. What does the normal jugular venous 'v' wave mainly represent?

A. Closure of the pulmonary semilunar valve.
B. Right atrial electrical depolarisation.
C. Completion of left ventricular ejection.
D. Atrial filling while tricuspid is closed.

43. What is detected by the fingers when palpating the radial pulse?

A. A direct pulse of venous blood into the radial artery.
B. An arterial wall-distension pressure wave.
C. Individual red cells striking the wrist.
D. An electrical signal passing along the endothelium.

44. V2 has been placed correctly and V4 is at its usual site. Where should V3 be attached?

A. At the fifth space, midaxillary line.
B. At the fifth space, anterior axillary line.
C. At the fourth space, right sternal border.
D. Halfway between the V2 and V4 electrodes.

45. Using Bazett's formula, QTc = QT/?RR, what is QTc when QT is 0.36 s and RR is 0.81 s?

A. 0.29 s, because QT is multiplied by RR.
B. 0.32 s, because QT is multiplied by ?RR.
C. 0.44 s, because RR is added to QT first.
D. 0.40 s, because QT is divided by 0.90.

46. Where is the largest controllable fall in systemic vascular pressure normally produced?

A. Across the venae cavae.
B. Within the right atrium alone.
C. Across resistance arterioles.
D. Across large elastic arteries.

47. Why does the sinoatrial node ordinarily set the heart's rhythm rather than latent pacemakers?

A. Its action potentials last longer than all other cells.
B. It has the fastest intrinsic depolarisation rate.
C. It receives oxygen from the pulmonary veins first.
D. Its cells have the largest ventricular muscle mass.

48. Which sequence correctly traces normal electrical conduction from atria into ventricular myocardium?

A. SA node -> Purkinje fibres -> AV node -> His bundle.
B. SA node -> AV node -> His bundle -> Purkinje fibres.
C. AV node -> SA node -> left ventricle -> right atrium.
D. Purkinje fibres -> atria -> SA node -> AV node.

49. Why can coordinated cardiac pumping continue without a new external nerve impulse for every beat?

A. The atrial myocardium conducts only during expiration.
B. Intrinsic pacemaker cells generate rhythmic impulses.
C. The sinoatrial node supplies the pressure needed for ejection.
D. Cardiac cells cannot spontaneously depolarise.

50. A patient has near-normal resting cardiac output but very high ventricular filling pressures and pulmonary congestion. Which statement fits best?

A. Heart failure can occur with normal output.
B. High filling pressure proves adequate pumping.
C. Heart failure requires reduced ejection fraction.
D. Normal resting output excludes heart failure.

51. What event ends isovolumetric contraction and begins ventricular ejection?

A. Ventricular pressure falls below atrial pressure.
B. Atrial contraction briefly increases end-diastolic volume.
C. Ventricular pressure falls below aortic pressure.
D. Ventricular pressure exceeds the outflow artery pressure.

52. Which pressure difference primarily drives blood back to the right atrium from the body?

A. Aortic systolic minus right ventricular systolic pressure.
B. Left ventricular systolic minus aortic diastolic pressure.
C. Systemic filling pressure minus right atrial pressure.
D. Pulmonary venous minus pulmonary capillary pressure.

53. What can follow when the right ventricle fails to deliver adequate forward flow to the lungs?

A. Pulmonary flow rises despite low right ventricular output.
B. Less blood reaches the left heart for systemic output.
C. Systemic tissues receive more oxygen-rich blood immediately.
D. Left ventricular filling necessarily increases sharply.

54. Where does the ductus arteriosus carry much fetal right ventricular output?

A. Directly from right atrium to left atrium.
B. From pulmonary trunk into descending aorta.
C. From left ventricle into hepatic portal vein.
D. From umbilical vein into inferior vena cava.

55. At an otherwise unchanged capillary, how would increased interstitial protein concentration affect fluid movement?

A. It increases protein attraction into the blood compartment.
B. It strengthens plasma oncotic pressure, reducing filtration.
C. It lowers capillary hydrostatic pressure, opposing filtration.
D. It strengthens the outward oncotic force favouring filtration.

56. Electrical stimulation of the medullary pressor area increases arterial pressure. What is the principal neural pathway?

A. Direct stimulation of the carotid sinus sensory fibres.
B. Inhibition of sympathetic neurons in the spinal cord.
C. Direct inhibition of firing in the sinoatrial pacemaker.
D. Excitation of sympathetic neurons in the spinal cord.

57. Which pair correctly identifies the lung's respiratory and supporting arterial circulations?

A. Bronchial arteries for gas exchange and coronary arteries for support.
B. Pulmonary arteries for support and hepatic arteries for gas exchange.
C. Pulmonary arteries: gas exchange; bronchial arteries: support.
D. Pulmonary veins for gas exchange and aortic veins for support.

58. A patient has low circulating volume. Which hormone chiefly increases renal water retention and can constrict vessels at high concentration?

A. Vasopressin, also called ADH.
B. Atrial natriuretic peptide.
C. Vasoactive intestinal peptide.
D. Endothelial prostacyclin.

59. Why might severe anaemia produce a cardiac flow murmur despite structurally normal valves?

A. Lower viscosity and increased flow favour turbulence.
B. Reduced blood volume necessarily closes the aortic valve.
C. Anaemia markedly increases blood viscosity and suppresses flow.
D. Anaemia blocks the ventricular conduction system.

60. During ventricular diastole, what would an increase in peripheral resistance do to the rate of aortic pressure decline if compliance were unchanged?

A. Speed the decline by increasing downstream runoff.
B. Leave the decline unaffected by downstream resistance.
C. Slow the decline by reducing downstream runoff.
D. Reverse arterial flow from capillaries toward the aorta.